Protocol-Specific Anomaly Detection

Packet Parser Based Anomaly Detection Project — Research Reference

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1. Quality of Service (QoS)

Common Anomalies

Anomaly	Description
Latency / Jitter Spikes	Sudden large increases in round-trip time or jitter, indicating congestion or DoS attacks
DSCP / Priority Misuse	Malicious hosts marking traffic with high-priority DSCP values improperly
Packet Loss / Throughput Drops	Anomalous packet loss rates or throughput fluctuations signaling network issues
SLA Violations	QoS policy misconfiguration causing priority inversion (low-priority traffic crowding out high-priority flows)
Buffer Overflow Events	Forwarding queue saturation in routers/switches
Bandwidth Degradation	Throughput falling below SLA thresholds

Detection Techniques

- **Threshold-based alerting** on metric baselines using Tukey's IQR method for outlier bounds
- **EWMA / CUSUM** on packet loss and throughput metrics for detecting gradual drift
- **Time-series anomaly detection** with change-point detection for baseline shifts
- **Deep learning** to detect DSCP–packet-size mismatches by comparing observed DSCP distributions to normal profiles
- **In-band Network Telemetry (INT)** — embedding telemetry directly into live packets for real-time visibility

- **SHAP-enhanced explainability** — integrating change-point detection, k-means clustering with Dynamic Time Warping, and supervised classifiers trained on QoS telemetry

ML Algorithms

Algorithm	Use Case
LSTM / LSTM Autoencoder	Time-series of RTT, jitter, loss; detects temporal anomalies
CNN-LSTM Hybrid	Captures spatial + temporal patterns in multi-metric QoS telemetry; 98.67% accuracy on UNSW-NB15
Random Forest / XGBoost	Strong supervised baseline for QoS metric classification
Self-Organizing Maps (SOM)	Unsupervised clustering of QoS workload patterns
Change-Point Detection	Detects abrupt shifts in jitter/latency baselines

Datasets

Dataset	Description	Link
UNSW-NB15	Multi-class attack + normal with QoS-relevant features	https://research.unsw.edu.au/projects/unsw-nb15-dataset
Network-Anomaly-Dataset (Kaggle)	QoS metrics: throughput, congestion, packet loss, latency, jitter	https://www.kaggle.com
CICIDS 2017 / 2018	Realistic labeled traffic with DoS/DDoS impacting QoS	https://www.unb.ca/cic/datasets

2. VoIP / RTP Traffic

Common Anomalies

Anomaly	Description

SIP BYE DoS	RTP media continues after a SIP BYE, indicating attack or misbehaving endpoint
RTP Flooding	Many RTP packets with duplicate SSRC or out-of-order sequence numbers
SIP INVITE/BYE/CANCEL Flood	Excess SIP messages per second (DoS) or many short-lived call attempts
Registration Hijacking	Multiple REGISTER requests for one user; track failed authentication attempts
SPIT (Spam over Internet Telephony)	Unsolicited call spam from spoofed VoIP accounts; high call frequency with low establishment ratio
RTP Stream Injection	Inserting forged packets into a live media session
One-Ring SPAM	Caller initiates then cancels before answer, luring callback to premium numbers
Codec Anomalies	Unexpected payload type changes mid-stream, indicating tampering

Detection Techniques

- **Stateful flow analysis** — track Call-ID, session start/stop, SIP state machines for illegal transitions (CANCEL before INVITE)
- **Rate limiting** on SIP methods (INVITE, REGISTER, BYE) per source IP
- **IPFIX/flow-based monitoring** — using IETF IPFIX templates tailored for SIP/RTP to detect BYE DDoS and RTP flooding
- **Hidden Markov Models (HMM)** — classify normal vs. malicious call state sequences
- **Deep Convolutional Autoencoder (DCAE)** — trained on normal VoIP traffic; achieves F1 scores of 99.32% and 99.56% on INRIA and RIT datasets respectively

ML Algorithms

Algorithm	Use Case
Deep Convolutional Autoencoder (1D-DCAE)	Best-performing unsupervised model for SPIT / RTP flooding detection
Hidden Markov Models (HMM)	Classify normal vs. malicious SIP call patterns
One-Class SVM	Trained on normal call patterns to flag outlier sessions

LSTM / GRU	Models temporal call flow sequences for behavioral anomalies
Random Forest / SVM	Supervised classification on labeled VoIP attack features
Isolation Forest	Unsupervised outlier detection on RTP/SIP flow features

Datasets

Dataset	Description	Link
RIT Enterprise VoIP Dataset	PCAPs of SIP/RTP under real attack conditions	https://www.mdpi.com/2076-3417/13/12/6974
INRIA VoIP Dataset	SIP messages including SPIT; 63,584 messages	Available via INRIA research
CAIDA PCAP Traces	Background traffic for mixing with VoIP attack scenarios	https://www.caida.org

3. Multicast Traffic

Common Anomalies

Anomaly	Description
IGMP / MLD Floods	Excessive IGMPv2/v3 or MLD join/leave messages causing multicast storms
Unauthorized Multicast	Packets sent to multicast groups from non-member hosts; PIM anomalies
Unusual Group Sizes	Sudden spike in receivers on a group, or one host joining many groups simultaneously
IGMP Spoofing	Forged group membership messages to gain unauthorized access to streams
PIM Manipulation	Forged PIM Hello messages disrupting multicast routing trees
Multicast Amplification DDoS	Using multicast addresses as reflectors for traffic amplification

Bandwidth Asymmetry	Abnormal upload/download ratio in multicast streams

Detection Techniques

- **IGMP message rate monitoring** per host; flag high-rate join/leave floods within a time window
- **Group address validation** — exclude reserved groups, flag non-standard multicast address usage
- **Entropy analysis** of group join patterns to detect scanning or enumeration
- **Tree topology verification** — validating multicast distribution tree consistency
- **Rate limiting + threshold detection** on IGMP control message bursts

ML Algorithms

Algorithm	Use Case
Isolation Forest	Effective on tabular IGMP/PIM flow features; fast and scalable
LOF (Local Outlier Factor)	Detects local density anomalies in join/leave behavior
DBSCAN	Clusters normal membership patterns; outliers flagged
Graph Neural Networks (GNN)	Models multicast tree topology; detects structural anomalies and unauthorized branches
Random Forest	Supervised classification of multicast attack types when labeled data is available
Autoencoder	Unsupervised baseline for normal multicast group behavior

Datasets

Dataset	Description	Link
CAIDA Anonymized Internet Traces	Contains multicast flows in backbone traffic	https://www.caida.org
Netresec Public PCAPs	Various PCAPs including multicast-relevant captures	https://www.netresec.com/?page=PcapFiles
CICIDS 2017 / 2018	Flooding traffic extractable for multicast-like behavior	https://www.unb.ca/cic/datasets

Custom GNS3 / EVE-NG Emulation	Best approach: simulate PIM/IGMP environments and generate labeled data	—

4. DHCP

Common Anomalies

Anomaly	Description
DHCP Starvation	Attacker sends many DHCPDISCOVER requests with spoofed MACs to exhaust the IP address pool (Gobbler attack)
Rogue DHCP Server	Unauthorized server responds to DISCOVER before legitimate server; multiple competing OFFER/ACK from non-authorized sources
DHCP Spoofing	Forged OFFER/ACK packets injecting malicious gateway or DNS server values
Duplicate IP Conflicts	Same IP leased to multiple hosts due to attack or misconfiguration
Abnormal Lease Renewal Rate	Unusually frequent REQUEST messages indicating misconfigured or malware-infected clients
DHCP Relay Anomalies	Unexpected relay agent IPs or Option 82 field manipulation

Detection Techniques

- **Rate-based detection** — counting DISCOVER/REQUEST messages per MAC per time window; flag bursts
- **MAC-to-IP binding verification** — flagging inconsistencies in DHCP binding tables
- **Authorized server whitelisting** — alerting on OFFER messages from unknown source IPs
- **Option field inspection** — validating DHCP option fields for malicious gateway or DNS injection
- **Autoencoder on DHCP flows** — dynamic IP address patterns handled by using only the first three octets; reconstruction error flags anomalies

ML Algorithms

Algorithm	Use Case

Autoencoder (DNN)	Learns normal DHCP lease patterns; flags reconstruction errors for unknown behavior
K-Means Clustering	Groups normal DHCP behavioral patterns; flags outlier clients
Random Forest	Classifies DHCP attack types (starvation vs. rogue server vs. spoofing)
LSTM / RNN	Models temporal DORA sequence (Discover → Offer → Request → Ack) for out-of-order anomalies
Threshold-based + Rule Engine	Simple but effective for rate-based starvation detection

Datasets

Dataset	Description	Link
ISOT-CID	Cloud intrusion dataset including DHCP-related anomalies	University of Victoria
CICIDS 2017	Contains flooding traffic adjacent to DHCP starvation patterns	https://www.unb.ca/cic/datasets
Netresec Public PCAPs	Includes DHCP captures for baseline analysis	https://www.netresec.com/?page=PcapFiles
Custom Yersinia / DHCPig	Tools for generating labeled DHCP attack traffic (starvation, rogue server)	—

5. DNS

Common Anomalies

Anomaly	Description
DNS Tunneling	Encoding data in DNS query/response fields (long, high-entropy subdomains) to exfiltrate data or bypass firewalls
DNS Amplification DDoS	High-volume queries with spoofed source to open resolvers using ANY records, amplifying attack traffic
DNS Cache Poisoning	Injecting forged responses to redirect traffic to attacker-controlled servers

Fast-Flux Domains	Malware C2 using rapidly changing A records (many IPs, very low TTL)
DGA (Domain Generation Algorithms)	Malware auto-generating random-looking domains to evade blocklists
DNS Hijacking	Unauthorized modification of DNS resolution paths
NXDOMAIN Flood	Many queries for non-existent domains to overwhelm resolvers
Subdomain Enumeration	Brute-force querying of subdomains as reconnaissance

Detection Techniques

- **Entropy analysis of domain names** — DGA and tunneling domains have high character-level entropy
- **TTL and A-record count analysis** — fast-flux domains show abnormally low TTL with many IPs per domain
- **Query frequency profiling** — beaconing malware queries the same domain at regular intervals
- **Response content validation** — checking answer sections against expected IP ranges
- **Deep Autoencoder profiling** — profiles DNS traffic into fixed-dimensional vectors; detects 170+ emerging malicious domains per deployment (Palo Alto Unit 42, 2024)
- **NXDOMAIN rate monitoring** — abnormally high non-existent domain response ratio

ML Algorithms

Algorithm	Use Case
Autoencoder (Deep)	Unsupervised profiling of DNS traffic per domain; high reconstruction error = anomaly
Stacked Ensemble (XGBoost + RF + OCSVM)	97.48% accuracy, 97.09% F1-score on multivariate DNS features
XGBoost / Random Forest	Supervised classification; strong on structured DNS flow features
One-Class SVM (OCSVM)	One-class learning on known-clean DNS traffic; detects DNS tampering
Character-level CNN / LSTM	Applied to raw domain name strings to detect DGA patterns

DBSCAN	Unsupervised clustering on domain features; outliers flagged

Datasets

Dataset	Description	Link
BCCC-CIC-Bell-DNS-2024	120+ DNS flow features; covers DDoS, amplification, exfiltration, hijacking, tunneling, poisoning	https://www.sciencedirect.com/science/article/abs/pii/S0045790624003641
CIC-Bell-DNS-2021 / EXF-2021	DNS exfiltration-focused dataset	University of New Brunswick
PUF Flow Dataset	DNS anomaly flows	https://shield-datasets.in/guest/projects/dataset-details-publicly/54
CTU-13 Dataset	Botnet C2 traffic including DNS-based communication	https://www.kaggle.com/datasets/dhoogla/ctu13
OONI Dataset	Two years of DNS tampering/censorship probes (2022–2023)	https://ooni.org/data/
Alexa Top 1M + DGA Samples	Benign vs. DGA domain name classification benchmark	—